One-Equation Turbulence Models

Problem

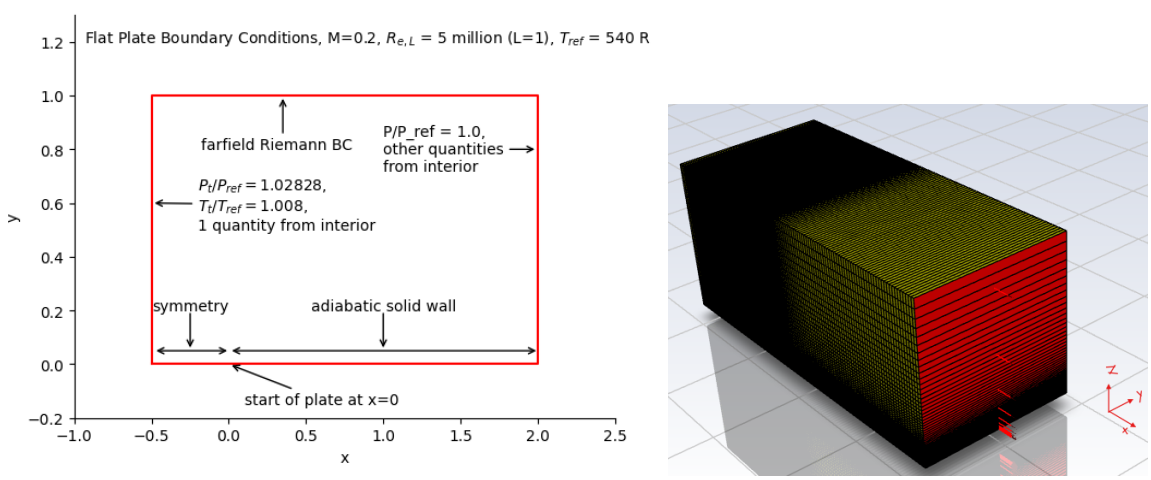
The objective of this study is to model the turbulent boundary layer over a 2D flat plate using a Computational Fluid Dynamics (CFD) solver ANSYS Fluent employing the one-equation (Spalart-Allmaras) model. This simulation will validate the CFD results against theoretical predictions by comparing:

- Wall skin friction coefficient (C_f) at different momentum-thickness Reynolds numbers ($4000 < R_{e,\theta} < 13000$).
- The law of the wall at $R_{e,\theta} = 10000$ to verify the near-wall velocity distribution.
- Eddy viscosity profiles at specific downstream locations (x=0.2, x=0.4, and x=0.6 m).
- Compare results with mixing-length model.

The aim is to evaluate the accuracy of the one equation model and its ability to capture essential flow characteristics in a turbulent boundary layer.

Solution

The study of boundary layer flow over a flat plate is fundamental in fluid mechanics, particularly in understanding transition from laminar to turbulent flow. Boundary conditions along with 2D representation of the plate is presented on the Figure 1a.



(a) 2D flat surface with boundary conditions.

(b) Mesh representation in Fluent.

Setup

At the inlet, a uniform free-stream velocity U_∞ is prescribed to simulate the flow approaching the flat plate. Turbulence intensity is set based on the Reynolds number, ensuring accurate boundary layer development. A pressure outlet boundary condition is applied at the downstream edge of the computational domain, with the pressure set to zero gauge pressure. This allows for a smooth exit of the fluid, reducing artificial reflections. The flat plate surface is modeled using a no-slip boundary condition, where both the streamwise and normal velocities are set to zero. This enforces the correct shear stress distribution necessary for boundary layer formation. A far-field boundary

condition is applied to the upper edge of the domain. This allows the flow to maintain free-stream conditions, with the velocity set to U_∞ and pressure set to ambient values.

A CFD solver is used to simulate the flow with the following settings:

- Turbulence Model: One-equation Spalart-Allmaras Model
- Grid: Structured mesh with refinement near the wall
- Solver: Pressure-based, steady-state
- Post-processing: Skin friction coefficient, the law of the wall and eddy viscosity plots

Set of equations used in post-processing is presented below.

$$Re_\theta = \frac{\rho_\infty U_\infty \theta}{\mu_\infty} \tag{1}$$

$$\theta = \int_0^\infty \frac{\rho}{\rho_\infty} \frac{u}{U_\infty} \left(1 - \frac{u}{U_\infty}\right) dy \tag{2}$$

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_\infty U_\infty^2} \tag{3}$$

$$\tau_w = \mu_w \left(\frac{\partial u}{\partial y}\right)_w \tag{4}$$

$$u^+ = \frac{u}{v^*} = \frac{u}{\sqrt{\tau_w/\rho_w}} \tag{5}$$

$$y^+ = \frac{y v^*}{\nu_w} = \frac{y\sqrt{\tau_w/\rho_w}}{\nu_w} \tag{6}$$

Spalart-Allmaras One-Equation Model

The transported variable in the Spalart-Allmaras model, $\widetilde{\nu}$, is identical to the turbulent kinematic viscosity except in the near-wall (viscosity-affected) region. The transport equation for the modified turbulent viscosity $\widetilde{\nu}$ is

$$\frac{\partial}{\partial t}\left(\rho\widetilde{\nu}\right) + \frac{\partial}{\partial x_i}\left(\rho\widetilde{\nu}u_i\right) = G_\nu + \frac{1}{\sigma_{\widetilde{\nu}}}\left[\frac{\partial}{\partial x_j}\left\{\left(\mu + \rho\widetilde{\nu}\right)\frac{\partial\widetilde{\nu}}{\partial x_j}\right\} + C_{b2}\rho\left(\frac{\partial\widetilde{\nu}}{\partial x_j}\right)^2\right] - Y_\nu + S_{\widetilde{\nu}}$$

where G_ν is the production of turbulent viscosity, and Y_ν is the destruction of turbulent viscosity that occurs in the near-wall region due to wall blocking and viscous damping. $\sigma_{\widetilde{\nu}}$ and C_{b2} are the constants and ν is the molecular kinematic viscosity. $S_{\widetilde{\nu}}$ is a user-defined source term. Note that since the turbulence kinetic energy, , is not calculated in the Spalart-Allmaras model.

The turbulent viscosity ν_t is computed from

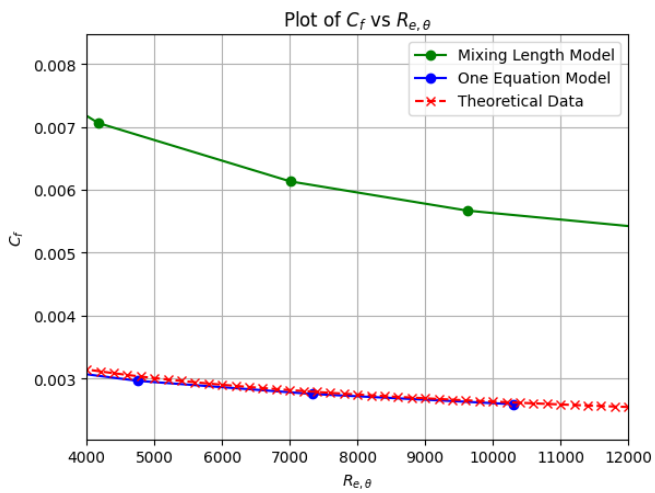
$$\nu_t = \rho\widetilde{\nu} f_{\nu 1} \tag{7}$$

$$f_{\nu 1} = \frac{\chi^3}{\chi^3 + C_{\nu 1}}, \qquad C_{\nu 1} = 0.71 \tag{8}$$

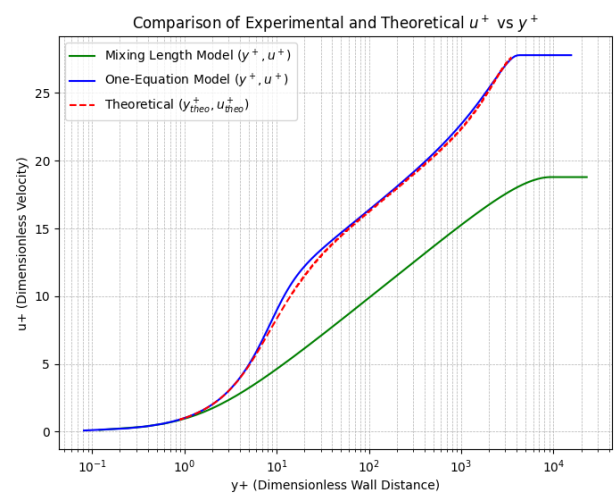
$$\chi \equiv \frac{\widetilde{\nu}}{\nu} \tag{9}$$

Results & Discussion

Distributions of the wall skin friction coefficient C_f over different Reynolds numbers against theoretical values for this case are presented on Fig. 2a. The Law of the Wall for different turbulence models against theoretical values is presented on the Fig. 2b.



(a) The wall skin friction coefficient over the Reylonds number for the range $4000 < R_e < 13000$ for one-equation and mixing length turbulence models.



(b) The law of the wall distributions for one-equation and mixing length turbulence models (R_e=10000).

One-Equation model results for C_f is showing better agreement with theoretical results across a range of Reynolds numbers than mixing length model, Fig. 2b. It captures the development of the turbulent boundary layer more accurately, especially at higher Reynolds number values. One-model equation can predict pressure gradients and flow separation better due to its ability to account for turbulent transport. The Spalart-Allmaras model is designed to resolve the near-wall region without needing a wall function.

The law of the wall is calculated and plotted against theoretical values for $R_{e,\theta} = 10000$. Result is depicted on Fig. 2b. In this case, closest Reynolds number of 10000 shows around surface line x=1.5. One-Equation Model more accurately reproduces the log-law region and viscous sublayer. It captures the full structure of the boundary layer, including the buffer and outer regions.

Eddy viscosity is plotted against the distance from the wall and for different surface positions. Results are presented on Fig. 3. In a viscous sublayer, the eddy viscosity $\nu_t \approx 0$, since the molecular viscosity dominates and the turbulence is negligible. After that, the mixing length gradually increases. The gradient of mean field velocity is still large, leading to rising eddy viscosity. The shape of ν_t begins to rise steeply. In log-law layer the ν_t reaches it maximum as some intermediate value y. Beyond this maximum, velocity gradient starts decreasing due to the slower velocity variation far from the wall. This causes the ν_t to gradually decrease in the outer layer. In the outer layer, the l saturates ar a value close to the boundary layer thickness. The velocity gradient continues decreasing and eddy viscosity starts decreasing.

Eddy viscosity distribution is plotted for three different stream-wise positions x=0.2, 0.4 and 0.6. It can be observed that for higher value of x, the eddy viscosity is larger, and it is because of the development of the turbulent boundary layer along the flow direction. As the fluid flows over a surface, the boundary layer thickens with increasing x. More turbulence is generated as the boundary layer grows, leading to higher turbulence intensity. Since eddy viscosity is

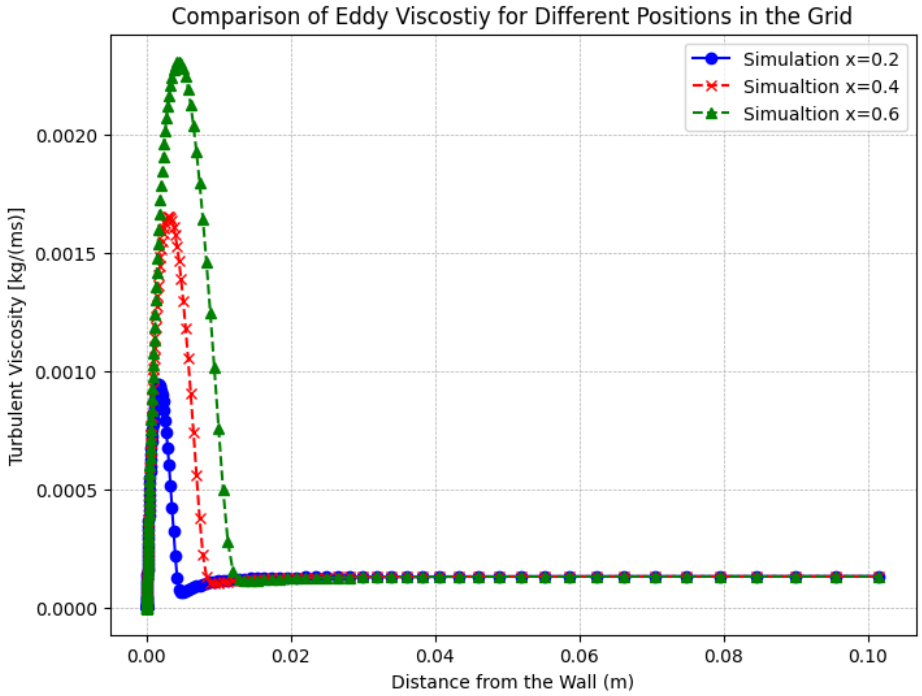


Figure 3: Eddy viscosity as a function of z-axis, distance from the wall for different positions at the surface x=0.2, 0.4 and 0.6, respectively.

proportional to turbulence intensity, it increases with x.

Conclusion

One-equation turbulence models, such as the Spalart-Allmaras model, provide an efficient approach for predicting turbulence in engineering applications, especially for flows where wall friction and boundary layer behavior are critical. These models solve a single transport equation, typically for eddy viscosity (ν_t), to represent turbulence effects. The eddy viscosity calculated by the model reflects the turbulence intensity near the wall, which directly influences C_f. The simplicity of the model and its calibration to experimental data ensure that it generally matches theoretical predictions for C_f, especially in fully-developed boundary layers.

These models capture the velocity profile accurately in the viscous sublayer and the logarithmic region. In particular, the Spalart-Allmaras model accurately represents the transition between these regions by adjusting the eddy viscosity based on the local turbulence characteristics. This enables the model to predict the velocity profile in close agreement with the theoretical law of the wall.

In the viscous sublayer, where molecular viscosity dominates, the eddy viscosity is very small (close to zero), reflecting the weak turbulence. As we move away from the wall into the log-law layer, the eddy viscosity increases, reaching a maximum before gradually decreasing in the outer region of the boundary layer. One-equation models like Spalart-Allmaras effectively capture these changes in eddy viscosity, which in turn influences the velocity gradients and turbulence intensity. The model's ability to adjust the turbulent viscosity properly leads to accurate predictions of C_f and the velocity profile.